Harder Geometrical Problems — problem 12.3.2.3

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In an equilateral convex hexagon $A_1A_2A_3A_4A_5A_6$, we have $\alpha_1 + \alpha_3 + \alpha_5 = \alpha_2 + \alpha_4 + \alpha_6$. Prove that $\alpha_1 = \alpha_4$, $\alpha_2 = \alpha_5$, $\alpha_3 = \alpha_6$ (α_i is the interior angle at vertex A_i).